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## Summary

ML/AI researcher and engineer in industrial R&D, with background spanning speech recognition, quantitative research, trading and machine learning systems. Experience building research platforms and production systems across forecasting, optimisation, and large-scale data-driven modelling.
*Now*: train & test -time reasoning in LLMs, applying ideas from ASR lattice decoding structured search hypothesis management. Agentic workflows, programmatic optimisation, auto- harness research & development, DSPy, meta-computational systems, automating the invention loop, agentic R&D as socratic dialogue for execution specification, inference as compute, context as memory, judging loops for evolving environments and synthetic data generation.
*Prior*: quantitative researcher, trader and portfolio manager developing systematic equity and FX models including low SNR Signal extraction, fitting, forecasting, portfolio optimisation, trading & risk management.
*Earlier*: PhD research in robust speech recognition in noise, MSc in text-to-speech synthesis and spoken document retrieval. Background in applied mathematics, statistics, computer science, and industrial R&D.

## Skills

Programming: C/C++/OpenMP, Python, SQL, duckdb, bash, awk, make, gdb/ddd, MATLAB, C#, R, Java.
Agents: omp/prime-agent/Hermes/pi/dsh/Codex/Claude; local LLM llama.cpp/strata/mlx-serve/hipfire.
Tools: Vim, Git, Github, tmux, VSCode, CLion, Jupyter, Spyder, Bloomberg & Reuters API-s.
Platforms: Linux (Ubuntu, CentOS), MacOS, MS-Windows, Gcloud, Slurm, HTCondor, Unix.

## Experience

**F9 Research, Director (2016–present; Harpenden, UK; remote distributed US, UK, EU)**
Rekindled ML/AI interests with OSS llama.cpp, open weights local LLMs (Qwen, Nemotron, DeepSeek, GLM); agents coding–omp/pi/prime-agent/Codex/OpenCode/Claude, general–Hermes/dsh, data–text2sql.
DNNs (TabM) for tabular data forecasting, TabPFN for regression (c.f. Hugging Face TabArena).
LLMs for earnings calls doc2vec features extraction for modelling and forecasting, reports reading.
Quant research & development short-horizons strategies using Python, C++, cluster & cloud resources.
Managed traded a market-neutral book gross ~$350M, trade ~$35M daily, EU & US markets (MATLAB).

**TalkMachine, Research Engineer (2026; remote distributed UK, EU, Dubai)**
LLM-s adaptations for a multilingual voice agent. LoRA to Qwen3-Omni and Nvidia-Nemotron-3.5-Transcribe. Curated and used Switchboard, AMI, ICSI corpora, created novel data using Qwen3.8-27B.

**FutureSearch, Research Scientist (2025; remote distributed US, UK, EU)**
Created then used agents to gather and organise financial data for end-user would-be products, and for internal use. Consulted on using the (presumed) alpha generated by the AI agent(s) for potential investment.

**Marshall Wace, Senior Quantitative Researcher (2010–2016; London, UK)**
Developed and scaled market-neutral portfolios from $100M to $10B+ over a period of 6 years.
Pioneered wrote unified R&D framework for data ingestion, signal extraction, modelling, portfolio optimization, simulation. Mentored junior researchers, implemented reproducible research workflows.

**Credit Suisse, Quantitative Analyst (2007–2009; London, UK)**
Independently traded ~35M gross equity market-neutral portfolios systematically, achieving 18% lifetime returns with Sharpe 3.1. Built and operated a complete trading platform for multi-market European equities.

**G-Research (DPFMG), Quantitative Analyst (2004–2007, London, UK)**
Designed and implemented systematic trading models for global equities and FX, contributed to fund profitability. Modelling, forecasting, risk management and multi-period optimization for mid- and high-frequency trading strategies. Operational portfolio management and production monitoring, on-call duty.

**Canon Research Europe, Researcher (2001-2004, Bracknell, UK)**
Embedded Automatic Speech Recognition, indexing, and retrieval of spoken documents with speech.

## Education

**Ph.D.** Computer Science – University of Sheffield, UK (2000)
Thesis: Robust Speech Recognition with Missing and Unreliable Data

**M.Phil.** Electrical Engineering – University Sv. Kiril i Metodij, Skopje, MK (1997)
Thesis: System for text-to-speech conversion for Macedonian language

**B.S.** Electrical Engineering – University Sv. Kiril i Metodij, Skopje, MK (1993)